mohankamath83@gmail.com
(770) 533-3913

Mohan Kamath

linkedin.com/in/mkamath
U.S. Citizen

Education

Georgia Institute of Technology — Expected May 2028
B.S. Computer Engineering — *Atlanta, GA*
GPA: 3.62/4.00
Relevant Coursework: Digital System Design, Circuit Analysis, Programming HW/SW Systems, Intro to Signal Processing, Fundamentals of Digital Signal Processing, Physical Foundations of Computer Engineering

Engineering Experience

SiliconJackets – Accelerator Group — Oct. 2026 – Present
Physical Design Engineer — *Atlanta, GA*

- Developing the physical implementation of the open-source Vortex GPU, targeting power and area efficiency for a 1-core, 4-warp, 4-thread configuration.
- Working on synthesis bring-up, toolchain integration, and SRAM sizing compatibility within the Vortex RTL-to-GDS physical-design flow.

VIP – Interactive Music Group — Aug. 2026 – Present
Undergraduate Researcher — *Atlanta, GA*

- Integrated an LSM6DSOX six-axis IMU with an ESP32-S3 over I2C; streamed timestamped sensor data in C/C++ and debugged sampling and serial-output bottlenecks.
- Implemented exponential moving average (first-order IIR) filtering of IMU data for smoother gesture control, addressing the tradeoff between noise suppression and control responsiveness.
- Mapped tilt and angular-rate inputs to note, octave, filter, and vibrato controls; integrated AMY audio synthesis with I2S output through the onboard codec.

Small Satellite Research Laboratory — Aug. 2025 – May 2026
Undergraduate Researcher — *Athens, GA*

- Developed software interfaces and telemetry serialization for the NASA-affiliated LEARNSat project; validated radio-linked data streams for mission simulations.

Selected Engineering Projects

- **RISC-V Processor Design & Verification** | *SiliconJackets Onboarding*
  - Implemented instruction fetch, arithmetic/logic execution, and SRAM access for an RV32I-style CPU in Verilog; passed all supplied digital-design tests.
  - Implemented driver, monitor, and scoreboard logic in a supplied SystemVerilog framework; verified CPU and memory behavior with directed tests, state-aware constrained-random stimulus, and assertions, achieving **98% DUT code coverage**.
- **Analog Audio Distortion Pedal** | *Op-Amps, Diode Clipping, Circuit Prototyping*
  - Prototyped an op-amp audio distortion circuit with adjustable voltage gain up to 101×.
  - Built hard-clipping diode networks and a voltage-divider bias circuit to establish a 4.5 V virtual ground and shape the audio waveform.
- **6-Key USB Mechanical Macropad** | *C, RP2350, TinyUSB, CMake*
  - Built RP2350 firmware with GPIO scanning, key debouncing, and TinyUSB HID.

Technical Skills

HDL & Verification: Verilog, SystemVerilog, SVA, Directed & Constrained-Random Testing
EDA Tools: Cadence Genus, Innovus, Virtuoso, Pegasus, Xcelium, SimVision, IMC; Synopsys Design Compiler
Physical Design: RTL Synthesis, Floorplanning, SRAM Macro Placement, Power Rings, Place-and-Route, Static Timing Analysis (WNS/TNS), Clock Gating, Cell Swapping, PVT Corners, Library Setup, DRC/LVS, SKY130
Programming: C/C++, Python, Tcl, RISC-V Assembly, MATLAB, Rust, TypeScript
Embedded Systems: ESP32-S3, RP2350, I2C, I2S, USB HID, TinyUSB
Hardware & Lab: KiCad, Onshape, Oscilloscope, Analog Discovery 3, Soldering
Development: Git, CMake, Linux/WSL; **DSP/Audio:** IIR Filtering, Audio Synthesis, Frequency Response